Discrete Random Variables

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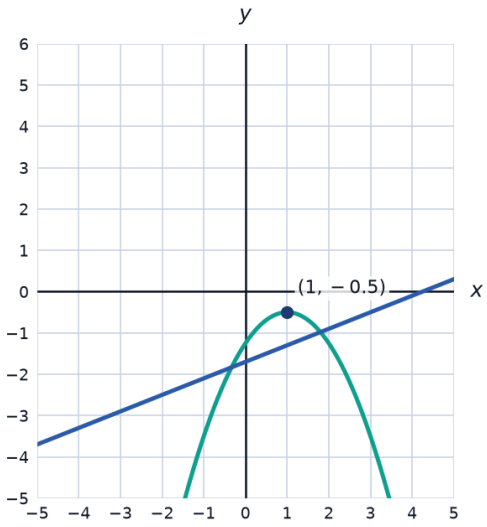
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Probability distributions**

A discrete random variable X has the probability distribution: $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, $P(X = 3) = k$, $P(X = 4) = 0.1$. Find k, using the fact that probabilities must sum to 1.

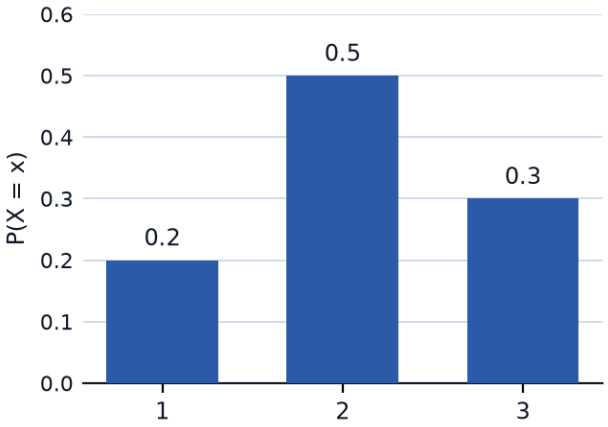
2 **[Foundational]** • **Target: Probability distributions**

A discrete random variable Y has the probability distribution: $P(Y = 0) = 0.1$, $P(Y = 1) = k$, $P(Y = 2) = 2k$, $P(Y = 3) = 0.3$. Find k.



3 **[Foundational]** • **Target: Expected value (mean) of a discrete random variable**

A random variable X has distribution $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. Find $E(X)$.



4 **[Foundational]** • **Target: Variance of a discrete random variable**

Using the distribution from the previous question ($E(X) = 2.1$), find $\text{Var}(X)$, using $\text{Var}(X) = E(X^2) - [E(X)]^2$.

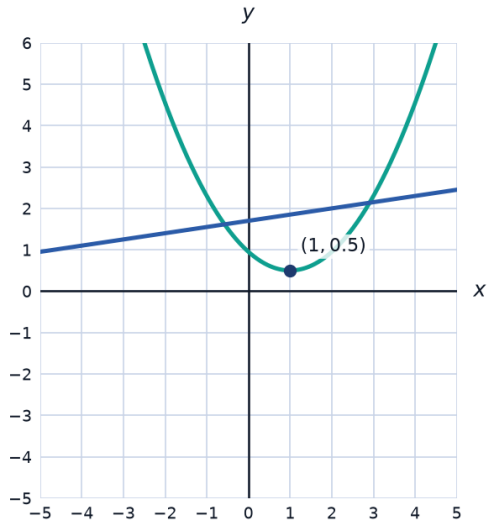
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5 [Foundational] • Target: Variance of a discrete random variable

Using the distribution $P(Y=0)=0.1$, $P(Y=1)=0.4$, $P(Y=2)=0.5$ ($E(Y)=1.4$), find $\text{Var}(Y)$, using $\text{Var}(Y)=E(Y^2)-[E(Y)]^2$.

6 [Foundational] • Target: Linear transformations of a random variable

A random variable X has $E(X)=5$ and $\text{Var}(X)=3$. Find $E(2X+4)$ and $\text{Var}(2X+4)$.



7 [Foundational] • Target: The discrete uniform distribution

A fair six-sided die is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, given that for a discrete uniform distribution on $1, \ldots, n$: $E(X)=\frac{n+1}{2}$ and $\text{Var}(X)=\frac{n^2-1}{12}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The discrete uniform distribution

A fair eight-sided die (labeled 1 to 8) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

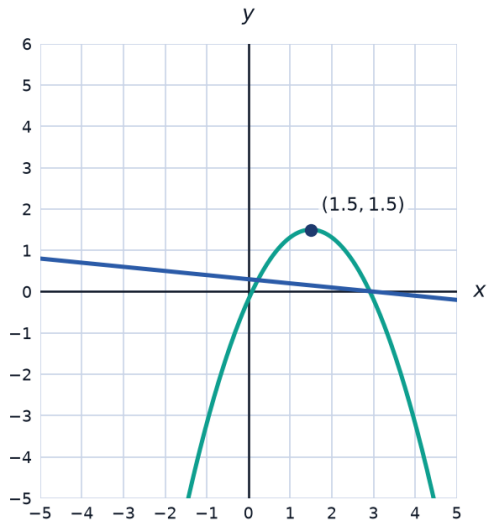
9 [Intermediate] • Target: The discrete uniform distribution

A fair ten-sided die (labeled 1 to 10) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

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10 [Intermediate] • Target: Constructing a probability distribution from a real scenario

A game involves rolling two fair dice and finding the sum X. Find $P(X = 4)$ and $P(X = 7)$.



11 [Intermediate] • Target: Constructing a probability distribution from a real scenario

Using the same game of rolling two fair dice and finding the sum X, find $P(X = 9)$ and $P(X = 12)$.

12 [Intermediate] • Target: A well-designed word problem: a raffle ticket game

This question has two parts. A raffle sells 200 tickets for 5 dollars each. One ticket wins a prize of 300 dollars, and two tickets each win a prize of 50 dollars. Let X be the net profit (prize won minus the 5-dollar ticket cost) for a single ticket. (a) Write the probability distribution of X. (b) Find $E(X)$, and interpret whether buying a ticket is a fair game.

13 [Intermediate] • Target: The discrete uniform distribution

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X = 5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

14 [Intermediate] • Target: Constructing a probability distribution from a real scenario

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

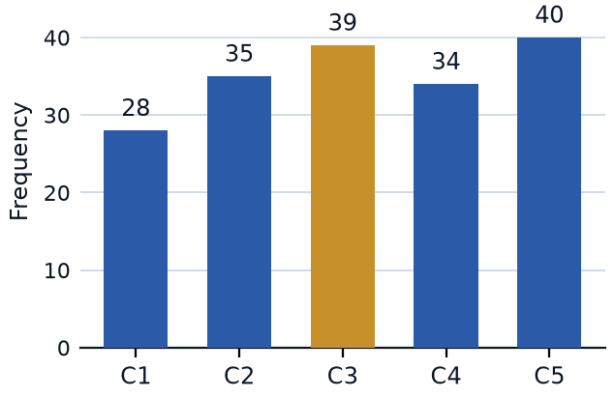
15 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 13.01 Newton's Laws

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \leq x \leq 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

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16 [Challenging] • Target: A well-designed word problem: a raffle ticket game • Cross-Lesson: 14.01 Energy, Work & Power

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.



17 [Challenging] • Target: Probability distributions • Cross-Lesson: 15.01 Representation of Data

The lifespan X in hours of an electronic component is normally distributed with mean $\mu = 1200$ hours and standard deviation $\sigma = 150$ hours. (a) Find the probability that a randomly selected component lasts more than 1350 hours. (b) In a batch of 500 components, estimate how many will last between 1000 and 1400 hours. (c) The manufacturer guarantees the top 10% longest-lasting components. Determine the minimum lifespan required to receive this guarantee.

18 [Challenging] • Target: Expected value (mean) of a discrete random variable • Cross-Lesson: 16.01 Permutations & Combinations

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

19 [Challenging] • Target: Variance of a discrete random variable • Cross-Lesson: 18.01 Binomial Distribution

Practice 2.2: The graph of $y = f(x)$ undergoes three consecutive transformations: (1) a horizontal stretch by scale factor 3, (2) a reflection in the y-axis, and (3) a translation by vector $(4;\ -5)$. Write down the equation of the resulting function $g(x)$ in terms of f.

20 [Challenging] • Target: Linear transformations of a random variable • Cross-Lesson: 19.01 Normal Distribution

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Probability distributions

A discrete random variable X is uniformly distributed on $\{1, 2, \ldots, 7\}$. Find $E(X)$.

a. 9/2

b. 3

c. 4

d. 5

22 [Foundational] • Target: Expected value (mean) of a discrete random variable

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. 0.980

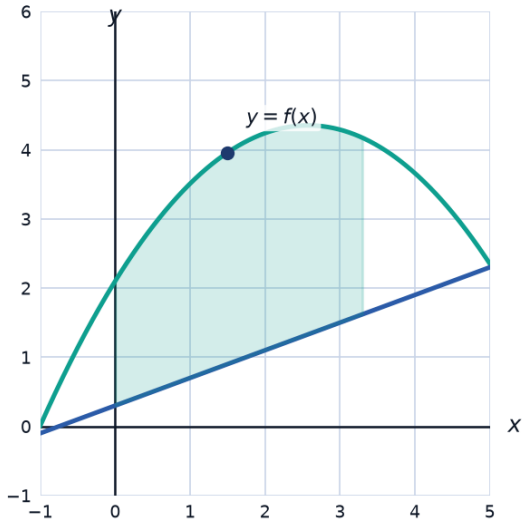
b. 0.900

c. 0.810

d. 0.990

23 [Foundational] • Target: Variance of a discrete random variable

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.



a. $\frac{127}{351}$

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

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24 [Foundational] • Target: Linear transformations of a random variable

The continuous random variable X has probability density function given by $f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise}, \text{ where } k \text{ is a constant. Find the value of } k.$

a. $\frac{3}{16}$

b. $\frac{1}{4}$

c. $\frac{3}{8}$

d. $\frac{16}{3}$

25 [Foundational] • Target: The discrete uniform distribution

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $\mathrm{P}(1 \le X \le 2)$.

a. $\frac{7}{8}$

b. $\frac{1}{8}$

c. $\frac{3}{4}$

d. $\frac{1}{2}$

26 [Foundational] • Target: Constructing a probability distribution from a real scenario

The continuous random variable X has probability density function $f(x)$ defined by \[f(x) = \begin{cases} kx & 0 \le x 1 \mid X < 2.5) \).

a. $\frac{9}{11}$

b. $\frac{10}{11}$

c. $\frac{3}{4}$

d. $\frac{9}{10}$

27 [Foundational] • Target: Cumulative distribution function

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

a. 6

b. 36

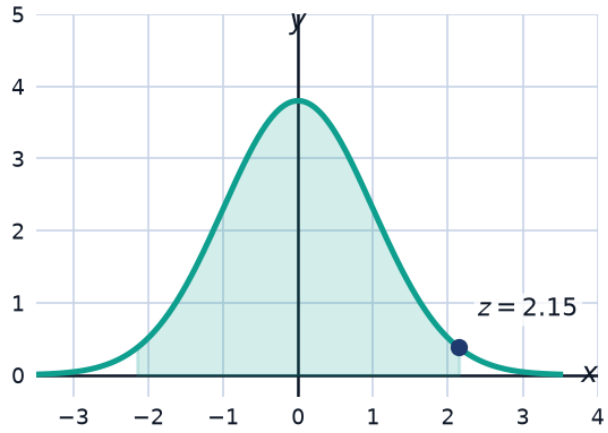
c. 12

d. $\sqrt{6}$

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28 [Foundational] • Target: A well-designed word problem: a raffle ticket game

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.



a. 44

b. 43

c. 11

d. 22

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Probability distributions

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

a. $\frac{19}{27}$

b. $\frac{8}{27}$

c. $\frac{7}{8}$

d. $\frac{2}{3}$

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30 [Intermediate] • Target: Expected value (mean) of a discrete random variable

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $P(X = 3) = P(X = 4)$, find the value of λ.

a. 4

b. 3

c. 12

d. $\frac{4}{3}$

31 [Intermediate] • Target: Variance of a discrete random variable

A random sample of 10 values of a variable x is taken. The results are summarised as follows: $\sum x = 50, \quad \sum x^2 = 322$ Calculate an unbiased estimate of the population variance.

a. 8

b. 7.2

c. 2.83

d. 32.2

32 [Intermediate] • Target: Linear transformations of a random variable

A continuous random variable X has PDF $f(x) = \frac{2}{2^2}x$ for $0 \le x \le 2$. Find the median of X in exact form.

a. $\frac{2\sqrt{2}}{2}$

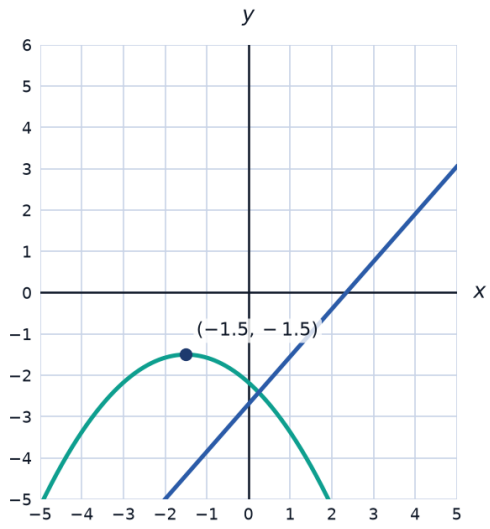
b. $\frac{2\sqrt{2}}{4}$

c. $\frac{2}{2}$

d. $2\sqrt{2}$

33 [Intermediate] • Target: The discrete uniform distribution

A continuous random variable X has PDF $f(x) = kx^1$ for $0 \le x \le 4$ (and 0 otherwise). Find k.



a. $\frac{5}{24}$

b. $\frac{1}{4}$

c. $\frac{1}{16}$

d. $\frac{1}{8}$

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34 **[Intermediate]** • **Target: Constructing a probability distribution from a real scenario**

A continuous random variable X has PDF $f(x) = \frac{2}{4^2}x^1$ for $0 \le x \le 4$. Find $\mathrm{Var}(X)$.

a. $\frac{17}{9}$

b. $\frac{8}{9}$

c. $\frac{8}{3}$

d. 8

35 **[Intermediate]** • **Target: Cumulative distribution function**

A binomial random variable has $n = 200$, $p = \frac{1}{50}$, and is approximated by a Poisson distribution. Find the parameter λ of the approximating Poisson distribution.

a. 50

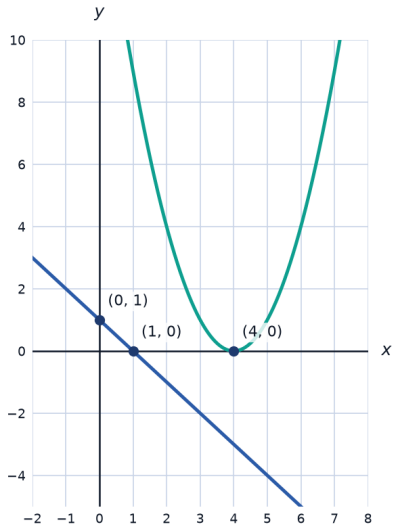
b. 4

c. 200

d. 5

36 **[Intermediate]** • **Target: A well-designed word problem: a raffle ticket game**

The graph of the quadratic function $f(x) = x^2 - 8x + k$ is tangent to the x-axis. The graph of the linear function $g(x) = -x + 1$ is also shown. Find the value of k.



a. 16

b. 8

c. 4

d. 2

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37 [Intermediate] • Target: Probability distributions

Find the distance between the points $(7, 4)$ and $(16, -8)$.

a. 21

b. 14

c. 15

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Expected value (mean) of a discrete random variable • Cross-Lesson: 16.01 Permutations & Combinations

The rational function f is defined by $f(x) = \frac{2x^4 - 3x^3 + kx^2 - 8x + 3}{(x-2)(x^2+1)}$, where k is a real constant. Given that the series expansion of $f(x)$ in ascending powers of x, valid for sufficiently small $|x|$, has no term in x^2, find the value of k.

a. $\frac{25}{4}$

b. $\frac{17}{4}$

c. $-\frac{25}{4}$

d. $\frac{115}{12}$

39 [Challenging] • Target: Variance of a discrete random variable • Cross-Lesson: 18.01 Binomial Distribution

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ or $5 < k < 11$

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

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40 [Challenging] • Target: Linear transformations of a random variable • Cross-Lesson: 19.01 Normal Distribution

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$

b. $16\mathrm{e}^{-4} \approx 0.293$

c. $4\mathrm{e}^{-2} \approx 0.541$

d. $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

41 [Challenging] • Target: The discrete uniform distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

a. $54\mathrm{e}^{-6} \approx 0.134$

b. $\frac{2}{3}\mathrm{e}^{-2} \approx 0.0902$

c. $\frac{2187}{8}\mathrm{e}^{-9} \approx 0.0337$

d. $\frac{625}{24}\mathrm{e}^{-5} \approx 0.175$

42 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 3.01 Graphs & Transformations

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3

b. 1

c. 3

d. 24

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43 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 4.01 Coordinate Geometry

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

a. $k = 5, \quad x_2 = \frac{28}{9}$

b. $k = 5, \quad x_2 = \frac{44}{9}$

c. $k = 2, \quad x_2 = \frac{14}{5}$

d. $k = 7, \quad x_2 = \frac{84}{25}$

44 [Challenging] • Target: A well-designed word problem: a raffle ticket game • Cross-Lesson: 5.01 Trigonometry

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$

b. $k = -2;\ x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

45 [Challenging] • Target: Probability distributions • Cross-Lesson: 6.01 Differentiation & Integration

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

a. The machine is likely calibrated correctly, as μ_0 is within the interval.

b. The machine is likely not calibrated correctly, as μ_0 is outside the interval.

c. The machine is likely calibrated correctly, as the sample mean is close to μ_0.

d. The machine is likely not calibrated correctly, as the sample standard deviation is too large.

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